

Problem

Find the charge $q(t)$ flowing through a device if the current is:

(a) $i(t) = 3 \text{ A}$, $q(0) = 1 \text{ C}$

(b) $i(t) = (2t + 5) \text{ mA}$, $q(0) = 0$

(c) $i(t) = 20 \cos(10t + \pi/6) \mu\text{A}$, $q(0) = 2 \mu\text{C}$

(d) $i(t) = 10 e^{-30t} \sin 40t \text{ A}$, $q(0) = 0$

Step-by-step solution

Step 1 of 6

Consider $i(t)$ is the current passing through a device.

Consider the equation for current $i(t)$ in terms of charge $q(t)$.

$$i(t) = \frac{d}{dt} q(t)$$

$$q(t) = \int_0^t i(t) dt + q(0) \dots\dots (1)$$

Here, $q(0)$ is the initial charge.

Step 2 of 6

(a)

The current is 3 A and initial charge is 1 C .

Substitute 3 A for $i(t)$ and 1 C for $q(0)$, in equation 1.

$$\begin{aligned} q(t) &= \int_0^t 3 + 1 \\ &= 3 \int_0^t 1 dt + 1 \\ &= 3t \Big|_0^t + 1 \\ &= 3(t - 0) + 1 \\ &= (3t + 1) \text{ C} \end{aligned}$$

Hence, for $i(t) = 3 \text{ A}$ and $q(0) = 1 \text{ C}$, the charge $q(t)$ is $\boxed{(3t + 1) \text{ C}}$.

Step 3 of 6

(b)

The current is $(2t+5)$ mA and initial charge is 0 C.

Substitute $(2t+5)$ mA for $i(t)$ and 0 C for $q(0)$, in equation (1).

$$\begin{aligned}q(t) &= \int_0^t ((2t+5) \times 10^{-3}) + 0 \\&= 10^{-3} \int_0^t (2t+5) \\&= 10^{-3} \left[(2) \frac{t^2}{2} \Big|_0^t + 5t \Big|_0^t \right] \\&= 10^{-3} ((t^2 - 0) + 5(t - 0)) \\&= 10^{-3} (t^2 + 5t) \\&= (t^2 + 5t) \text{ mC}\end{aligned}$$

Hence, for $i(t) = (2t+5)$ mA and $q(0) = 0$ C, the charge is $\boxed{(t^2 + 5t) \text{ mC}}$.

Step 4 of 6

(c)

The current is $20 \cos\left(10t + \frac{\pi}{6}\right)$ μ A and initial charge is 2 μ C.

Substitute $20 \cos\left(10t + \frac{\pi}{6}\right)$ μ A for $i(t)$ and 2 μ C for $q(0)$, in equation (1).

$$\begin{aligned}q(t) &= \int_0^t \left(20 \cos\left(10t + \frac{\pi}{6}\right) \right) \times 10^{-6} + (2 \times 10^{-6}) \\&= (20 \times 10^{-6}) \int_0^t \cos\left(10t + \frac{\pi}{6}\right) + (2 \times 10^{-6}) \\&= (20 \times 10^{-6}) \left[\sin\left(10t + \frac{\pi}{6}\right) \left(\frac{1}{10} \right) \right]_0^t + (2 \times 10^{-6}) \\&= (2 \times 10^{-6}) \left[\sin\left(10t + \frac{\pi}{6}\right) \right]_0^t + (2 \times 10^{-6}) \\&= (2 \times 10^{-6}) \left[\sin\left(10t + \frac{\pi}{6}\right) - \sin\left(0 + \frac{\pi}{6}\right) \right] + (2 \times 10^{-6}) \\&= (2 \times 10^{-6}) \left[\sin\left(10t + \frac{\pi}{6}\right) - \sin(30^\circ) \right] + (2 \times 10^{-6}) \\&= (2 \times 10^{-6}) \left[\sin\left(10t + \frac{\pi}{6}\right) - \frac{1}{2} \right] + (2 \times 10^{-6}) \\&= (2 \times 10^{-6}) \sin\left(10t + \frac{\pi}{6}\right) - (1 \times 10^{-6}) + (2 \times 10^{-6}) \\&= (2 \times 10^{-6}) \sin\left(10t + \frac{\pi}{6}\right) + (1 \times 10^{-6}) \\&= \left[2 \sin\left(10t + \frac{\pi}{6}\right) + 1 \right] \mu\text{C}\end{aligned}$$

Hence, for current $i(t) = 20 \cos\left(10t + \frac{\pi}{6}\right)$ μ A and $q(0) = 2$ μ C, the charge is

$$\boxed{\left[2 \sin\left(10t + \frac{\pi}{6}\right) + 1 \right] \mu\text{C}}.$$

Step 5 of 6

(d)

The current is $10e^{-30t} \sin(40t)$ A and initial charge is 0 C.

Substitute $10e^{-30t} \sin(40t)$ A for $i(t)$ and 0 C for $q(0)$, in equation (1).

$$q(t) = \int_0^t (10e^{-30t} \sin(40t)) dt \dots\dots (2)$$

Consider,

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx)) \dots\dots (3)$$

Step 6 of 6

Compare equation (2) and equation (3).

By comparing $a = -30$ and $b = 40$.

$$\begin{aligned} 10 \int_0^t (e^{-30t} \sin(40t)) dt &= 10 \left[\frac{e^{at}}{a^2 + b^2} (a \sin(bt) - b \cos(bt)) \right] \\ &= 10 \left[\frac{e^{-30t}}{30^2 + 40^2} (-30 \sin(40t) - 40 \cos(40t)) \right] \\ &= \frac{10}{2500} [e^{-30t} (-30 \sin(40t) - 40 \cos(40t))] \\ &= -e^{-30t} [0.16 \cos(40t) + 0.12 \sin(40t)] \text{ C} \end{aligned}$$

Hence, for $i(t) = 10e^{-30t} \sin(40t)$ A and $q(0) = 0$ C, the charge is

$$\boxed{-e^{-30t} [0.16 \cos(40t) + 0.12 \sin(40t)] \text{ C}}.$$